

Quick Check

Lesson 9: Consolidation and Digital Animations: Comparing Mitosis and Meiosis to Explain Growth

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 1.4: Cell Division | Lesson 9

A cell in a growing child's leg bone has 46 chromosomes. It divides by mitosis to help the child grow. How many chromosomes will each new cell have?

- A** 23, because chromosomes are halved in every division
- B** 92, because the new cells add up the chromosomes
- C** 69, because each new cell gets half of the extra chromosomes
- D** 46, because each new cell gets a full copy of the set

During mitosis, at which stage do the sister chromatids pull apart and move to opposite poles of the cell?

A Prophase

B Metaphase

C Telophase

D Anaphase

One diploid ($2n$) cell in the testes of a bull divides by meiosis. What is produced at the end?

- A** Four genetically different haploid cells
- B** Two genetically different haploid cells
- C** Four genetically identical diploid cells
- D** Two genetically identical diploid cells

A fertilised egg divides by mitosis. Every cell divides once in each round, and no cells die. How many cells will there be after 5 rounds of mitosis?

A 16 cells

B 25 cells

C 10 cells

D 32 cells

A learner says, 'A child grows because the ugali, beans and sukuma wiki we eat turn directly into new cells.' Which statement best corrects this idea?

- A** Food supplies the chromosomes that are put into each new cell
- B** Food supplies energy and raw materials, but new cells come from existing cells dividing by mitosis
- C** Food makes the existing cells bigger, and that is the only way the body grows
- D** Food is not needed because cells make everything they need by meiosis

A girl falls and cuts her knee, and the cut heals after a few days. Which process makes the new skin cells that close the wound, and why?

- A** Mitosis, because it makes identical diploid cells for growth and repair
- B** Meiosis, because it halves the chromosome number to make skin cells
- C** Mitosis, because it makes genetically different cells with crossing over
- D** Meiosis, because it makes four new cells from each cell

A group makes a model of meiosis in a cow. A cow body cell has 60 chromosomes. How many chromosomes should their model show in each egg cell (gamete) at the end of meiosis?

A 120 chromosomes

B 30 chromosomes

C 15 chromosomes

D 60 chromosomes

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).